

Dear Hiring Committee,

I am excited to apply for the Software Engineering Internship at Hudson River Trading for Summer 2026. I am a penultimate-year Artificial Intelligence and Computer Science student at the University of Edinburgh, currently on exchange at Carnegie Mellon University. With experience spanning large-scale systems, quantitative research, and applied machine learning, I am drawn to HRT's mission of applying scientific and technological innovation to build some of the most advanced trading infrastructure in the world.

In my most recent work at PlythosAI, I developed a large-scale conversational optimisation system that fine-tuned transformer-based models to enhance contextual reasoning and response efficiency. This project involved end-to-end system design, from building distributed data pipelines in Python and PyTorch to deploying models on cloud infrastructure optimised for low latency and scalability. Previously, as a research intern, I worked on classifying speculative execution vulnerabilities in low-level C and assembly code, which strengthened my understanding of performance optimisation, concurrency, and systems-level programming.

I've also built a foundation in algorithmic trading through participation in competitions such as Optiver's Coding Challenge, Citadel's Terminal, IMC's Prosperity Challenge, and the Carnegie Mellon Algorithmic Trading Competition. These experiences taught me how to think critically under time constraints, design efficient systems, and reason about code performance in real-world, high-pressure environments.

What excites me most about HRT is its culture of engineering excellence, curiosity, and collaboration, where writing clean, efficient, and elegant code directly powers real-world impact. My background in developing intelligent systems, tutoring courses in object-oriented and functional programming, and co-founding a quantitative modeling society has reinforced my belief that innovation thrives where creativity meets disciplined engineering which is very similar to HRT's values.

I would be thrilled at the opportunity to contribute to HRT's engineering teams by building performant and reliable systems that push the limits of what's possible in technology and trading. Thank you for considering my application, and I look forward to the opportunity to bring my experience and enthusiasm to Hudson River Trading.

Best regards,
Ritvik Gupta